

Least-Information Action on Delay Embeddings: Testing Rose–Takens with Local Linear Diffusion Kernels

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Abstract

1 Introduction

Thesis (applied). Choose coordinates and a local diffusion kernel that minimize the one-step KL *edge action* between data and model; when the action is ≈ 0 (the Rose condition), the reconstruction attains measure and spectral equivalence.

2 Setup and Principle

Let x_t evolve on an attractor A ; form a delay vector $y_t = \Psi_k(x_t) \in \mathbb{R}^{k+1}$. Optionally reduce with a linear basis $z_t = Ly_t \in \mathbb{R}^m$, $m \geq 2d + 1$. Let ν be the stationary law on $\mathcal{Y}$ (either Y or Z space).

Kernel and edge law. A one-step transition kernel $K(y_0, dy_1)$ induces the joint edge measure

$$\Gamma(dy_0, dy_1) = \nu(dy_0) K(y_0, dy_1). \quad (1)$$

Least-information action. Given a fitted model kernel $\hat{K}$, define the *edge action*

$$\mathcal{A}_{\text{edge}} = D_{\text{KL}}(\Gamma_{\text{data}} \parallel \Gamma_{\text{model}}) = D_{\text{KL}}(\nu(dy_0)K \parallel \nu(dy_0)\hat{K}). \quad (2)$$

The *Rose test* accepts when $\mathcal{A}_{\text{edge}} \approx 0$ (with uncertainty), evidencing kernel equality on support.

3 Methods

3.1 Basis (optional) and neighborhoods

We over-embed via Ψ_k and optionally choose a well-conditioned linear basis L (e.g., PCA-whiten Y_t then apply a small random rotation; or use a Fisher/CCA basis). Neighborhoods around a query state y_0 (or z_0) are built with kNN or kernel weights $w_t = w(\|y_t - y_0\|/h)$.

3.2 Local linear diffusion kernel

For a query y_0 , fit a locally weighted linear map

$$(A(y_0), b(y_0)) = \arg \min_{A, b} \sum_t w_t \|y_{t+1} - Ay_t - b\|^2, \quad \mu(y_0) = A(y_0) y_0 + b(y_0). \quad (3)$$

Residual covariance (ridge-stabilized):

$$\Sigma(y_0) = \frac{\sum_t w_t r_t r_t^\top}{\sum_t w_t} + \lambda I, \quad r_t = y_{t+1} - A(y_0) y_t - b(y_0). \quad (4)$$

Define the empirical kernel (single-Gaussian; mixture optional):

$$\hat{K}(y_0, dy_1) = \mathcal{N}(\mu(y_0), \Sigma(y_0)) dy_1. \quad (5)$$

3.3 Hyperparameters

Choose h (neighborhood scale), λ (ridge), and optional mixture size by maximizing held-out conditional log-likelihood $\sum \log \hat{K}(y_t, y_{t+1})$ (equivalently, minimizing forward KL).

3.4 Estimating the edge action (Rose test)

Use a time-blocked hold-out set $\mathcal{H}$ of edges to estimate

$$\hat{\mathcal{A}}_{\text{edge}} = \frac{1}{|\mathcal{H}|} \sum_{(y_0, y_1) \in \mathcal{H}} \log \frac{\hat{p}_{\text{emp}}(y_1 | y_0)}{\hat{K}(y_0, y_1)}, \quad (6)$$

where $\hat{p}_{\text{emp}}(\cdot | y_0)$ is a light-smoothing baseline (e.g., coarse bins or kNN) ensuring domination. Report a bootstrap CI; accept Rose if CI contains 0. Optionally report a symmetric proxy (JS divergence).

3.5 Optional: Spectral/geometry diagnostics

(i) Koopman correlation moments $c_k = \langle h, U^k h \rangle$ vs empirical; (ii) Diffusion maps from a symmetrized affinity $W(y_i, y_j) = \frac{1}{2} [\hat{K}(y_i, y_j) \nu(y_i) + \hat{K}(y_j, y_i) \nu(y_j)]$ to compare leading spectra.

4 Algorithms (concise)

Algorithm 1: FitLocalDiffusionKernel(y_0 , data, h , λ)

Compute weights $w_t = w(\|y_t - y_0\|/h)$.
Solve (??) for (A, b) ; set $\mu(y_0) = Ay_0 + b$.
Compute $\Sigma(y_0)$ via (??).
Return $\hat{K}(y_0, \cdot) = \mathcal{N}(\mu(y_0), \Sigma(y_0))$.

Algorithm 2: RoseTest(data, $\hat{K}$)

Hold out a time block $\mathcal{H}$; build $\hat{p}_{\text{emp}}(\cdot | y_0)$.
Compute $\hat{\mathcal{A}}_{\text{edge}}$ and a bootstrap CI.
Decision: accept Rose if CI contains 0.

5 Experiments

Datasets.

Protocol. Over-embed Ψ_k ; optional basis L ; time-blocked train/val/test; prevent leakage.

Baselines. Ulam counts (partition), kNN/KDE conditional.

Metrics. KL-action (and CI), predictive NLL, optional spectral/DMAPS checks.

6 Results

7 Discussion

8 Conclusion

Reproducibility.

A Least-Information Action (reference)

Path-space variant (optional): for a Markov model Q vs data/reference P ,

$$D_{\text{KL}}(\mathbb{P}_Q \| \mathbb{P}_P) = D_{\text{KL}}(Q_0 \| P_0) + \sum_{t=0}^{T-1} \mathbb{E}_Q \left[\log \frac{K_Q(Y_t, Y_{t+1})}{K_P(Y_t, Y_{t+1})} \right]. \quad (7)$$

Near a minimum, the Fisher–Rao quadratic supplies a local metric: $D_{\text{KL}}(\theta + \delta\theta \| \theta) = \frac{1}{2} \delta\theta^\top I(\theta) \delta\theta + o(\|\delta\theta\|^2)$.